

Senior Integration Hardening Scientific Data Analysis v2.7

Mother and children prepared for deliberate synchronization with Scientific
Workbench

Architectural health: canonical registry, multi-root ownership, accurate coverage claims,
stub integrity, and an app-side plan

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Translation of the preserved private v2.7 guide at commit d6583f1. Original TeX SHA-256:
9e10ef1a7e63f5e3f3ac6e41a0eb1d6d419f363fd8faf0dbb827ec10aef89c93. Private source:
skills/scientific-data-analysis/.cache/active_references/guides/guia_scientific_
data_analysis_v2_7.tex.

This is dated evidence, not current release verification. The original guide prepares an integration
handoff; it does not perform synchronization. Its recorded ratings and open app-side work retain
that historical meaning. The original remains unchanged privately.

Abstract

`scientific-data-analysis` v2.7 continues v2.6 with a different focus: addressing correctable risks identified by the senior AI / Agents / Plugins / Skills evaluation before resynchronizing the skill family with Scientific Workbench, rather than measuring local cost.

Mother and children already have a recorded 100/A `plugin-eval` rating. v2.7 adds the missing application contract: resolve compact canonical registries, treat mother and children as distinct roots, describe smoke/regression coverage accurately, verify symlink/stub integrity, and provide a short app-side synchronization plan.

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1. Executive overview

v2.7 is a senior hardening release. It creates no new capabilities, changes no scientific contracts, and does not edit Scientific Workbench. It prepares the mother and child skills for app consumption without fragile heuristics.

The product decisions are:

- `scientific-data-analysis` remains the normal entrypoint for broad tasks.
- `scientific-data-astro`, `scientific-data-documents`, `scientific-data-notebooks`, and `scientific-data-maintainer` are independent roots for app-side consumption.
- Scientific Workbench must resolve `canonical_registry` recursively.
- Every capability must execute from its owning root, rather than always from the mother.
- Current coverage is smoke/regression coverage, not exhaustive coverage.
- Maintainer remains maintainer-only and must not appear as an ordinary action.

2. From v2.6 to v2.7

Area	v2.7 change
Senior evaluation	Uses the senior evaluation dated July 1, 2026.
P1 app-side	Documents canonical registry resolution and multi-root ownership.
P2 coverage	Documents that <code>coverage-summary.json</code> describes smoke/regression coverage, not exhaustive coverage.
P2 integrity	Adds regression checks for broken symlinks, unresolved stubs, and incomplete canonical registries.
P2 drift	Freezes the app-side handoff to reduce drift between exposure/app-readiness and Swift code.
Concise documentation	Adds <code>v2-7-scientificworkbench-sync-plan.md</code> as a short guide for the app task.
Guide	This guide replaces v2.6 as the active document.

3. Applied corrections

Senior finding	Status	Correction
Resolve <code>canonical_registry</code>	PASS	Recursive regression reaches YAML with 61 entries.
Do not assume one root	PASS skill-side / WARNING app-side	Multi-root contract documented; app synchronization remains pending.
Non-exhaustive coverage	PASS	Artifacts declare smoke/regression coverage, not exhaustive coverage.
Symlinks/stubs/wrappers	PASS	Regression checks for broken symlinks in <code>references/</code> .
Exposure/app-readiness drift	Documentation PASS	Handoff requires app drift tests before synchronization is declared.
Dense documentation	PASS	Short plan dedicated to Scientific Workbench.

4. Multi-root contract for Scientific Workbench

The app must load these five installed roots:

Owner	Installed root
mother	<code>~/.codex/skills/scientific-data-analysis</code>
astro	<code>~/.codex/skills/scientific-data-astro</code>
documents	<code>~/.codex/skills/scientific-data-documents</code>
notebooks	<code>~/.codex/skills/scientific-data-notebooks</code>
maintainer	<code>~/.codex/skills/scientific-data-maintainer</code>

The mother may remain the visible entrypoint for broad tasks, but command construction must use each capability's owning root as `workingDirectory`. This prevents the app from consuming an incomplete or outdated system view.

5. Canonical registry contract

Each `public_surface_registry.yaml` may be compact. A consumer must:

1. open the root's registry;
2. resolve `canonical_registry` relative to the stub when present;
3. repeat until reaching YAML with `entries::`;
4. verify the expected 61-entry surface;
5. retain a fingerprint/cache without rewriting the skill.

6. v2.7 regression

New regression:

```
python scripts/audit_v2_7_senior_integration_regression.py
```

It should return `PASS` with an accepted warning if Scientific Workbench still uses a single `DefaultPaths.skillRoot`. That warning is not a skill failure; it identifies the work to complete during app synchronization.

7. Recommended gates

```
python3 -m py_compile scripts/audit_v2_7_senior_integration_regression.py
python scripts/audit_v2_7_senior_integration_regression.py
python scripts/audit_v2_7_plugin_eval_coverage_artifacts.py
python scripts/audit_v2_6_benchmark_matrix_regression.py
python scripts/audit_v2_5_budget_deferral_regression.py
python scripts/sync_public_surface_docs.py --check
python scripts/portable_smoke_test.py --profile core
```

In maintainer, the complete regression lives in `fixtures/`; the mother retains a compatible wrapper.

8. Short Scientific Workbench synchronization plan

App-side synchronization must follow this order:

1. Verify the installed skill using the v2.7 gates.
2. Create `SkillRootCatalog` or its equivalent in the app.
3. Resolve `canonical_registry` recursively for each root.
4. Associate every `CapabilityEntry` with its owning root.
5. Update `CapabilityCommandBuilder` to execute from the owning root.
6. Keep `ToolEnvelopeParser` and `ArtifactDiscovery` compatible with v1.8, v1.9, and v2.x.
7. Hide maintainer-only actions in normal mode.
8. Run Swift tests, the quality gate, and smoke cases for tables, documents, notebooks, FITS, and an absent optional backend.

9. Non-goals

- Do not synchronize Scientific Workbench during this phase.
- Do not copy skills into the app bundle.
- Do not turn maintainer into an ordinary user action.
- Do not present smoke/regression coverage as exhaustive.
- Do not change correct scientific behavior to satisfy the app.

10. Conclusion

v2.7 prepares the skill family with a recorded automatic 100/A rating, approved senior evaluation, closed skill-side P1/P2 corrections, and a concrete plan for Scientific Workbench to stop depending on a single root. The recorded next step is deliberate application synchronization using this contract.